Growfinix

Artificial Intelligence

INTERNSHP

TASK

(Complete 5/3 Task)

2-Month Internship 5 Practical AI Use Case Tasks

Presented By:
Growfinix Team

Appy Internships
https://www.growfinix.in

Growfinix

TASK 1

Focus: Real Estate Price Predictor (Regression Model)

Real Estate Price Predictor Description

1. Objective: Build a machine learning model to predict house prices.
2. Dataset: Use a CSV file with features like area, location, bedrooms, etc.
3. Data Cleaning: Handle missing values and convert categorical data using encoding.
4. Feature Selection: Choose key inputs like area, number of rooms, and location.
5. Model Type: Use Linear Regression from sklearn.linear_model.
6. Train & Test: Split the data into training and testing sets using train_test_split.

Tech Stack

Python, scikit-learn, Pandas, Matplotlib/Seaborn.

Visualization

Plot predictions vs actual prices to visualize accuracy.

Model Training

Fit the model and evaluate performance using mean_squared_error

Goal

Learn how regression models work and how to apply them to real-world data.

Growfinix

TASK 2

Email Spam Detector (NLP + Classification)

1. Objective: Build a machine learning model to classify emails as spam or not.
2. Text Preprocessing: Clean the text (lowercase, remove punctuation, stopwords, etc.).
3. Feature Extraction: Convert text to numbers using CountVectorizer or TfidfVectorizer.
4. Model Selection: Use classifiers like Naive Bayes or Logistic Regression.
5. Train/Test Split: Divide data into training and testing sets using train_test_split.
6. Prediction: Test the model on new messages to check spam detection.

Tech Stack

Python, scikit-learn, Pandas, and Natural Language Processing (NLP).

Dataset

Use a labeled dataset containing spam and non-spam messages.

Model Training

Fit the classifier on training data and evaluate accuracy.

Goal

Understand how machines process and classify human language using NLP.

Growfinix

TASK 3

Movie Recommendation System

1. Objective: Recommend movies to users based on their past ratings.
2. Dataset: Use a movie ratings dataset with user IDs, movie IDs, and ratings.
3. Approach: Apply collaborative filtering to find similar users or movies.
4. Data Preparation: Create a user-movie matrix from the dataset.
5. Similarity Check: Use cosine similarity to find users with similar tastes.
6. Prediction: Recommend movies that similar users liked

   but the current user hasn't seen.

Tech Stack

Python, Pandas, scikit-learn, and optionally surprise or lightfm.

Evaluation

Use metrics like RMSE or precision to check accuracy.

Optional UI

Display top 5 movie recommendations for a selected user ID.

Goal

Learn how AI can personalize experiences using user behavior data.

https://www.growfinix.in

Growfinix

TASK 4

Controlled Assistant

1. Objective: Create a voice-controlled assistant that understands and responds to basic commands.
2. Speech Input: Use speech_recognition to capture and convert voice to text.
3. Voice Output: Use pyttsx3 to make the assistant speak responses.
4. Basic Commands: Handle tasks like telling time, opening websites, or greeting the user.
5. If-Else Logic: Use simple conditional statements to process spoken commands.
6. Text Fallback: Allow manual text input if voice fails.

Tech Stack

Python, speech_recognition, pyttsx3, and datetime

Offline Capability

Works without internet using local speech engines.

No AI Model Needed

Purely rule-based – ideal for beginners in voice programming.

Goal

Learn how to build interactive Python apps using voice input and output.

https://www.growfinix.in

Growfinix

TASK 5

AI Research Task

1. Objective: Explore how AI is transforming a specific industry with real examples.
2. Research Focus: Study current AI applications, tools, and technologies in that field.
3. Real-World Impact: Include examples like AI in cancer detection, fraud detection, or crop monitoring.
4. Benefits: Highlight how AI improves speed, accuracy, and decision-making.
5. Challenges: Mention ethical issues, data privacy, or job impact if relevant.
6. Sources: Use news articles, case studies, or research papers for information.

Field Selection

Choose one area like healthcare, finance, or agriculture.

Presentation

Prepare a 1-page PDF or a short PowerPoint to present your findings.

Language

Keep the explanation simple and beginner-friendly.

Goal

Understand the real-life role of AI and improve your research and writing skills.

https://www.growfinix.in

Growfinix

THANK YOU

FOR YOUR ATTENTION

This internship program is thoughtfully designed to provide students with a structured and hands-on learning experience in Artificial Intelligence.

+91 9405256144

support@growfinix.in